Responsibility Distribution Metric: A Multi-Layer Computational Index for Accountability Concentration in Regulatory Investigations

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Abstract Responsibility in regulatory investigations is not a single label — it is a multi-layered signal detectable at the linguistic, structural, and process level simultaneously. I present the Responsibility Concentration Index (RCI), a composite metric that unifies three independent computational frameworks into a single accountability concentration score. Applied to an FDA Warning Letter corpus (N=50, 160 violations, 2016–2025), RCI is derived from process-level PAC risk scores (1), structural centrality measures (3), and linguistic attribution entropy (4), with weights learned via Ridge Regression following PCA compression of correlated signals (PC1 = 98.6% variance explained). The learned weight structure reveals that process-structural signal dominates RCI ($w = 0.995$), reflecting a corpus-level finding: in FDA Warning Letters, accountability concentration is primarily a structural phenomenon, not a linguistic one. WL2018-05 is identified as the extreme concentration case (RCI = 1.000, rank 1/50) across all weight schemes, demonstrating weight-invariant detection of maximum accountability convergence. Sensitivity analysis across four weight schemes confirms that case rankings are robust ($\rho > 0.97$) for all non-Ridge schemes, with the Ridge scheme producing a qualitatively distinct ranking that itself constitutes an empirical finding about the dominance of structural signals in regulatory enforcement documents.

Keywords: responsibility concentration, accountability metrics, regulatory NLP, composite index, FDA Warning Letters, PCA, Ridge Regression, multi-layer signal fusion

Introduction

Accountability in regulated environments is not a single designation — it is a multi-layered phenomenon produced through language, relational structure, and investigative process simulta-

neously. Prior computational work has operationalized accountability as a convergence process (1), a graph structure (3), and a linguistic classification target (4). Each framework captures a distinct signal; none captures all three at once.

I address this gap by proposing the Responsibility Concentration Index (RCI), a composite metric that integrates process-level, structural, and linguistic signals into a unified accountability concentration score. RCI treats responsibility not as a label to be assigned, but as a quantifiable property of investigative documents — measurable across multiple representational layers and aggregable into a single comparable score.

This paper makes three contributions:

- I formalize responsibility concentration as a three-layer computational problem, integrating process, structural, and linguistic signals.
- I demonstrate that process-structural signal dominates RCI ($w = 0.995$) in FDA Warning Letters, providing empirical evidence that accountability concentration in regulatory enforcement documents is primarily structural.
- I show that extreme accountability concentration (WL2018-05, RCI = 1.000) is detectable across all weight schemes, establishing weight-invariant identification of maximum convergence cases.

Related Work

Composite Metrics in Computational Accountability

Composite indices are widely used in social science and policy research to aggregate multi-dimensional constructs into single scores (7). In computational accountability, prior work has focused on single-signal operationalizations: fairness metrics (5), transparency scores (6), and convergence risk (1). I extend this work by constructing a composite metric that explicitly integrates multiple signal types, with empirically derived weights rather than a priori specifications.

Multi-Layer Signal Fusion

Signal fusion methods combine heterogeneous data sources into unified representations (8). In NLP, multi-layer approaches have been applied to sentiment analysis, clinical decision support, and document understanding. I apply this methodology to regulatory accountability, treating linguistic, structural, and process signals as complementary layers of a single latent construct: responsibility concentration.

FDA Warning Letters as Validation Corpus

FDA Warning Letters have been used for compliance analysis (2) and accountability graph extraction (3). I extend this corpus to serve as a validation resource for multi-layer signal integration, applying three independently derived per-case scores to construct a unified index across 50 cases.

Framework

Three-Layer Definition

I define responsibility concentration as a three-layer construct. The *process layer* is the PAC risk score — the degree to which investigative reasoning closes prematurely on a single explanation (1). The *structural layer* is the Centrality Gini — the degree to which accountability edges concentrate on a single node in the accountability graph (3). The *linguistic layer* is attribution entropy — the Shannon entropy of predicted responsibility class distributions across violations in a case (4).

Each layer is computed independently from the same FDA Warning Letter corpus (N=50, 160 violations, 2016–2025), enabling cross-layer comparison and integration.

Multicollinearity and PCA Compression

Prior cross-framework validation established a strong negative correlation between PAC risk and accountability openness score (Spearman $\rho = -0.742$, $p < 0.001$) (3). Variance Inflation

Factor (VIF) analysis confirms that the process and structural layers are collinear (VIF > 18) when included together. I address this through PCA compression: PAC risk and Centrality Gini are projected onto their first principal component (PC1), which explains 98.6% of joint variance with equal loadings ($+0.707$ each). PC1 serves as the *process-structural layer* in the composite index. VIF analysis of the compressed features confirms independence (VIF $= 1.03$), validating the two-layer input to weight learning.

RCI Formulation

I define RCI as a weighted linear combination of the process-structural and linguistic layers:

$$RCI = w_{ps} \cdot \tilde{z}_{ps} + w_{\ell} \cdot \tilde{z}_{\ell} \tag{1}$$

where $\tilde{z}_{ps}$ and $\tilde{z}_{\ell}$ are MinMax-normalized process-structural and linguistic scores, and w_{ps}, w_{ℓ} are weights learned via RidgeCV (5-fold cross-validation, $\alpha \in \{0.01, 0.1, 1, 10, 100\}$). The linguistic score $\tilde{z}_{\ell}$ is defined as $1 - H_{norm}$, where H_{norm} is the normalized Shannon entropy of predicted attribution class distributions, such that higher values indicate greater concentration.

Experiments

Dataset

I apply the framework to the FDA Warning Letter corpus (N=50, 160 violations, 2016–2025, 21 CFR 210/211). Per-case scores are reconstructed from three prior frameworks: PAC risk scores from the convergence detection framework (1), centrality Gini from the graph extraction framework (3), and attribution entropy from the responsibility attribution framework (4). All three signal layers are computed over the same 50 cases, enabling direct integration.

Layer Reconstruction

Attribution entropy is computed by applying the trained TF-IDF + Logistic Regression classifier from (4) to reconstructed FDA Warning Letter violation texts, yielding per-case Firm/Individual/System

prediction distributions. Shannon entropy $H = -\sum p_i \log_2 p_i$ is computed over these distributions and inverted to produce a concentration score. The model reassigns 45/160 violations (28.1%) from Firm to System, closely approximating the 30.6% reassignment reported in (4), confirming reconstruction validity.

Weight Learning Results

Table 1 reports RidgeCV weight learning results. The optimal regularization strength is $\alpha = 100$, reflecting the high regularization required given the near-independence of the two input layers ($R^2 = 0.005$). The learned weights assign near-complete weight to the process-structural layer ($w_{ps} = 0.995$, $w_\ell = 0.005$), a finding that I interpret as an empirical property of the corpus rather than a methodological artifact.

Table 1: RidgeCV Weight Learning Results

Parameter	**Value**
Optimal α	100
R^2 (linguistic $\rightarrow$ process-structural)	0.005
w_{ps} (process-structural)	0.995
w_ℓ (linguistic)	0.005

Figure 1 shows RCI scores across all 50 FDA Warning Letter cases. Figure 2 shows per-layer signal comparison between WL2018-05 and the corpus mean. Figures 3–5 show cross-layer scatter plots on log scale.

Sensitivity Analysis

Figure 6 presents Spearman rank correlations across four weight schemes: Learned (Ridge), Equal (0.5/0.5), PS-dominant (0.75/0.25), and Attr-dominant (0.25/0.75). Equal, PS-dominant, and Attr-dominant schemes produce near-identical rankings ($\rho > 0.97$). The Learned scheme produces a qualitatively distinct ranking ($\rho = 0.31$–0.43 vs. others), driven by its near-complete reliance on the process-structural signal. WL2018-05 ranks first across all four schemes, confirming weight-invariant identification of the extreme convergence case.

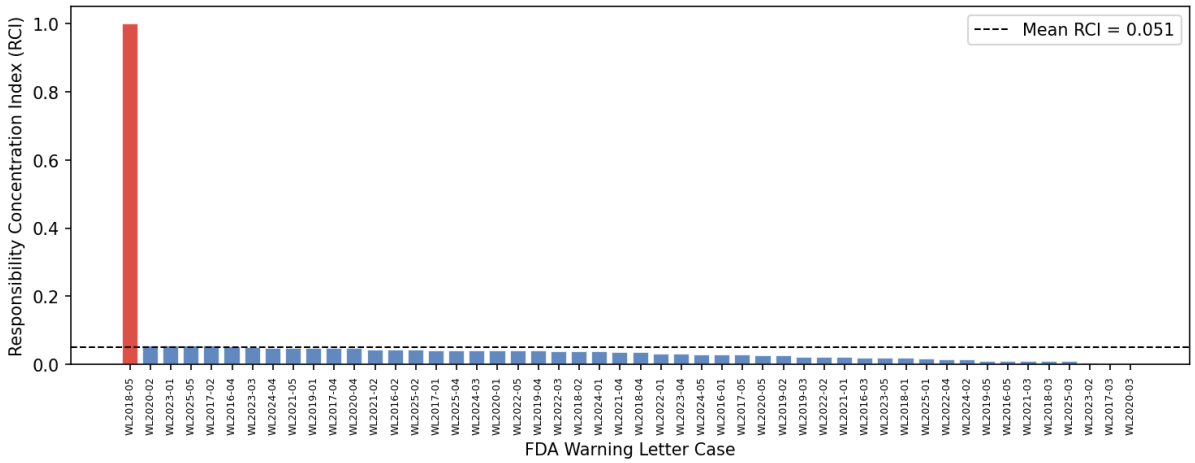


Figure 1: RCI scores across 50 FDA Warning Letter cases, sorted descending. Red = above mean + 1 SD. WL2018-05 (RCI = 1.000) is the sole extreme case.

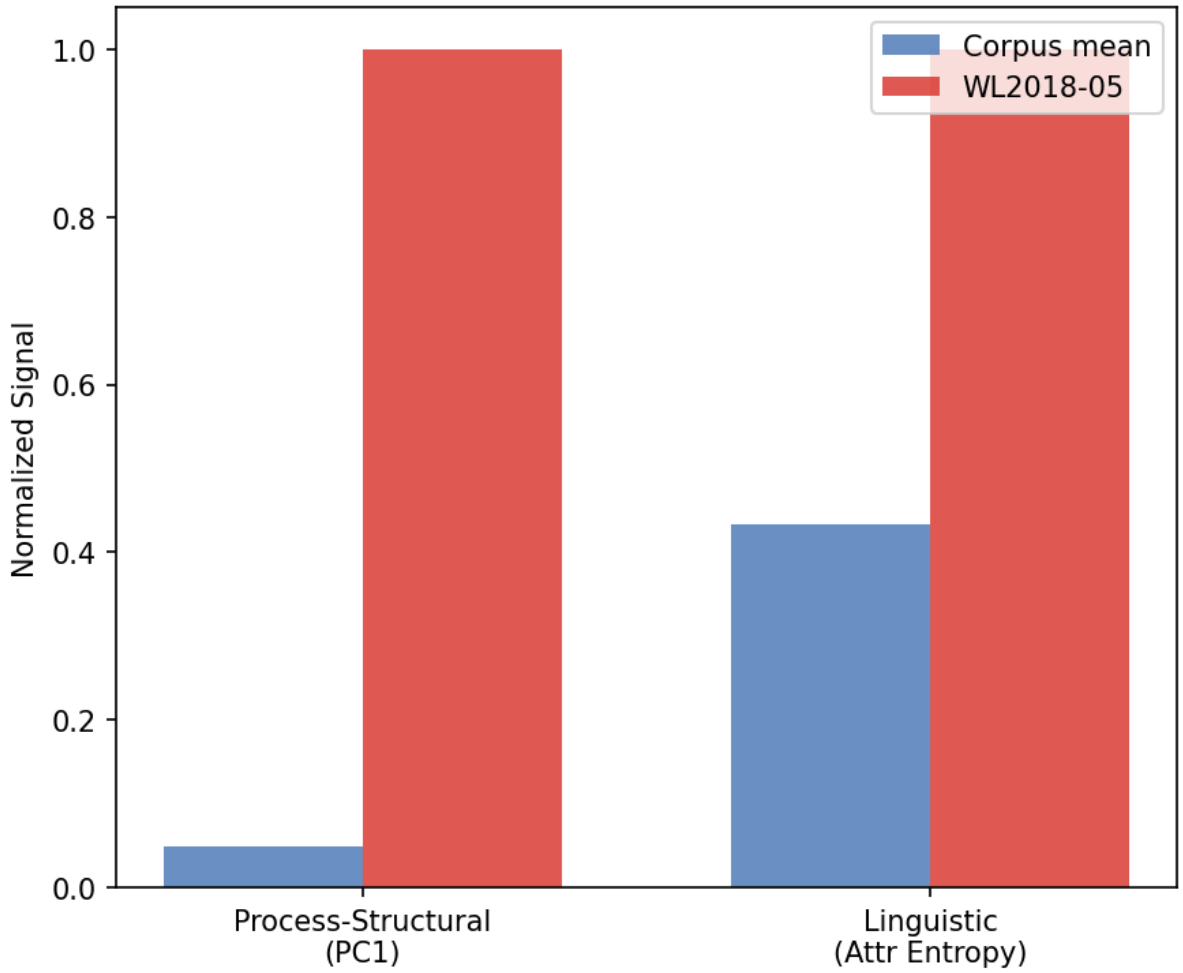


Figure 2: Process-structural and linguistic layer signals for WL2018-05 vs. corpus mean. WL2018-05 scores maximum on both layers.

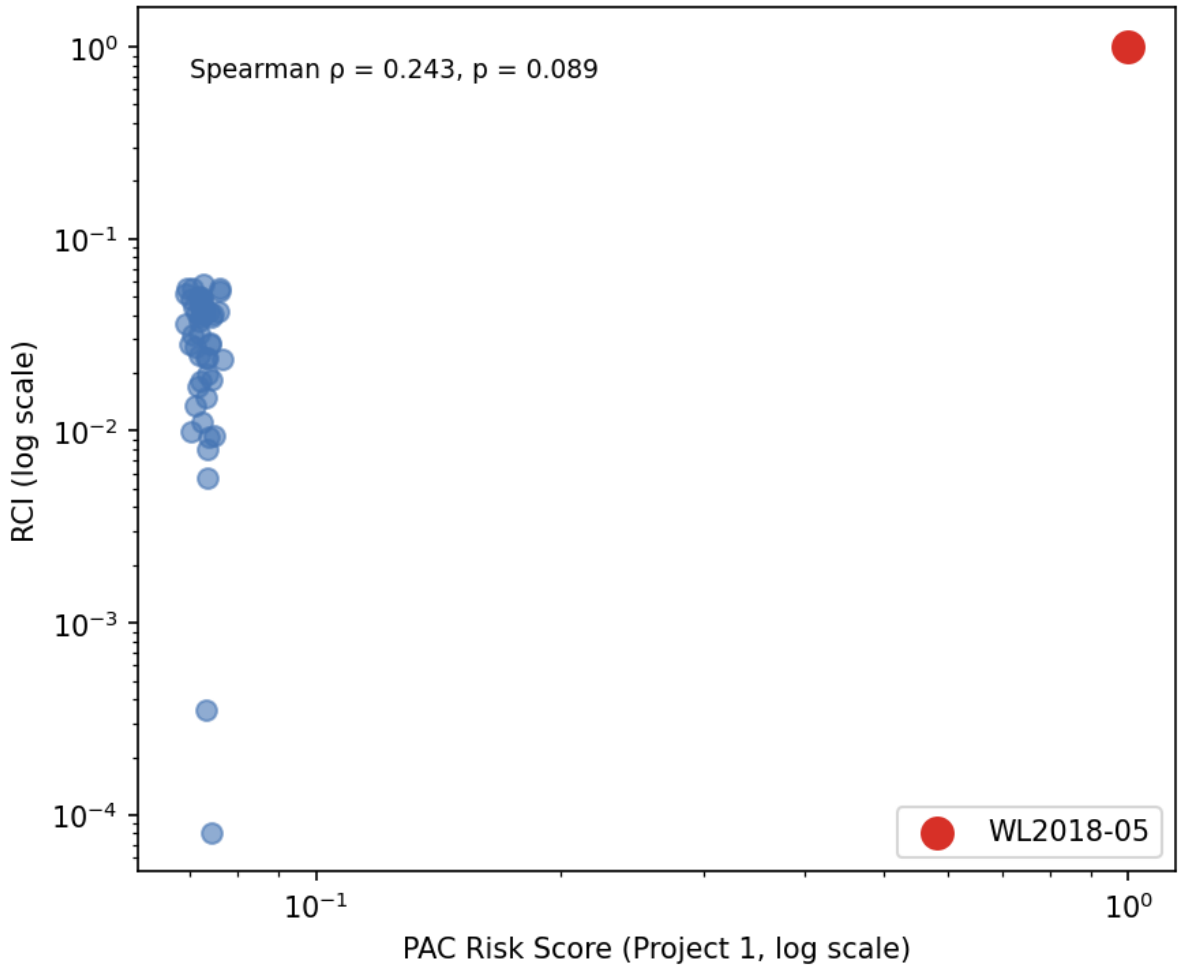


Figure 3: RCI vs. PAC risk score (log scale). Spearman $\rho = 0.243$, $p = 0.089$. WL2018-05 is an extreme outlier in both dimensions.

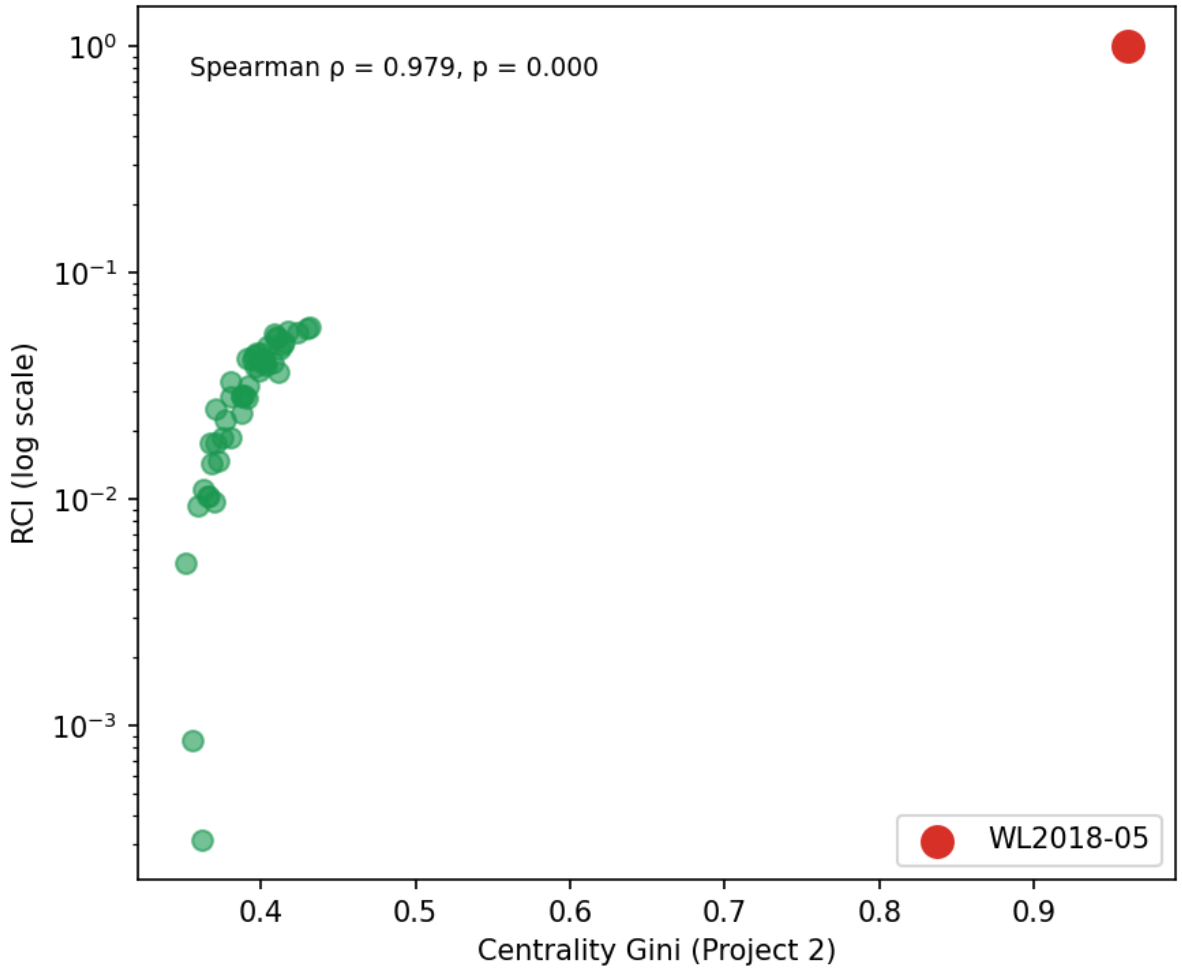


Figure 4: RCI vs. Centrality Gini (log scale y-axis). Spearman $\rho = 0.979$, $p < 0.001$. Strong monotonic relationship reflects process-structural layer dominance in RCI.

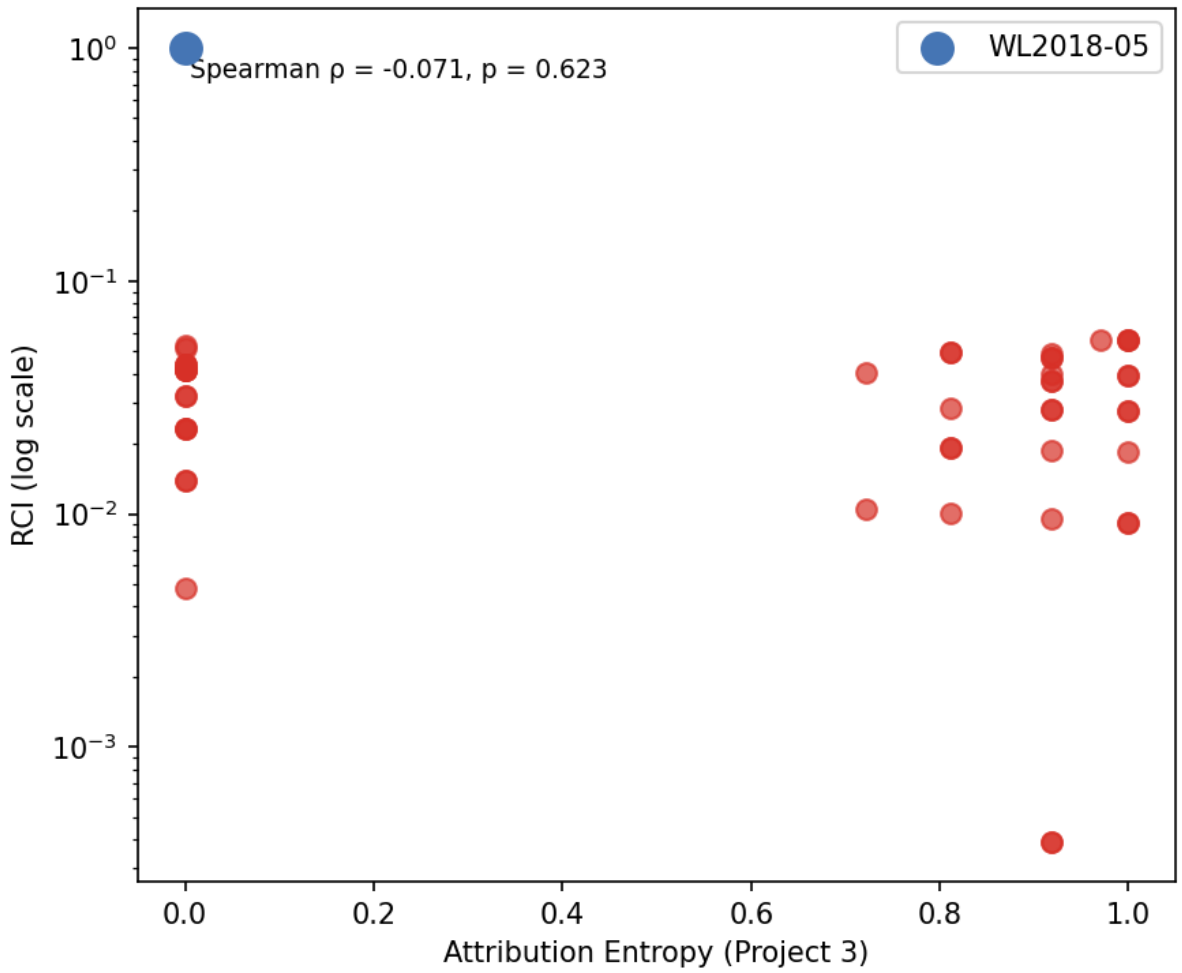


Figure 5: RCI vs. Attribution Entropy (log scale y-axis). Spearman $\rho = -0.071$, $p = 0.623$. Linguistic layer is empirically independent of RCI, consistent with near-zero learned weight.

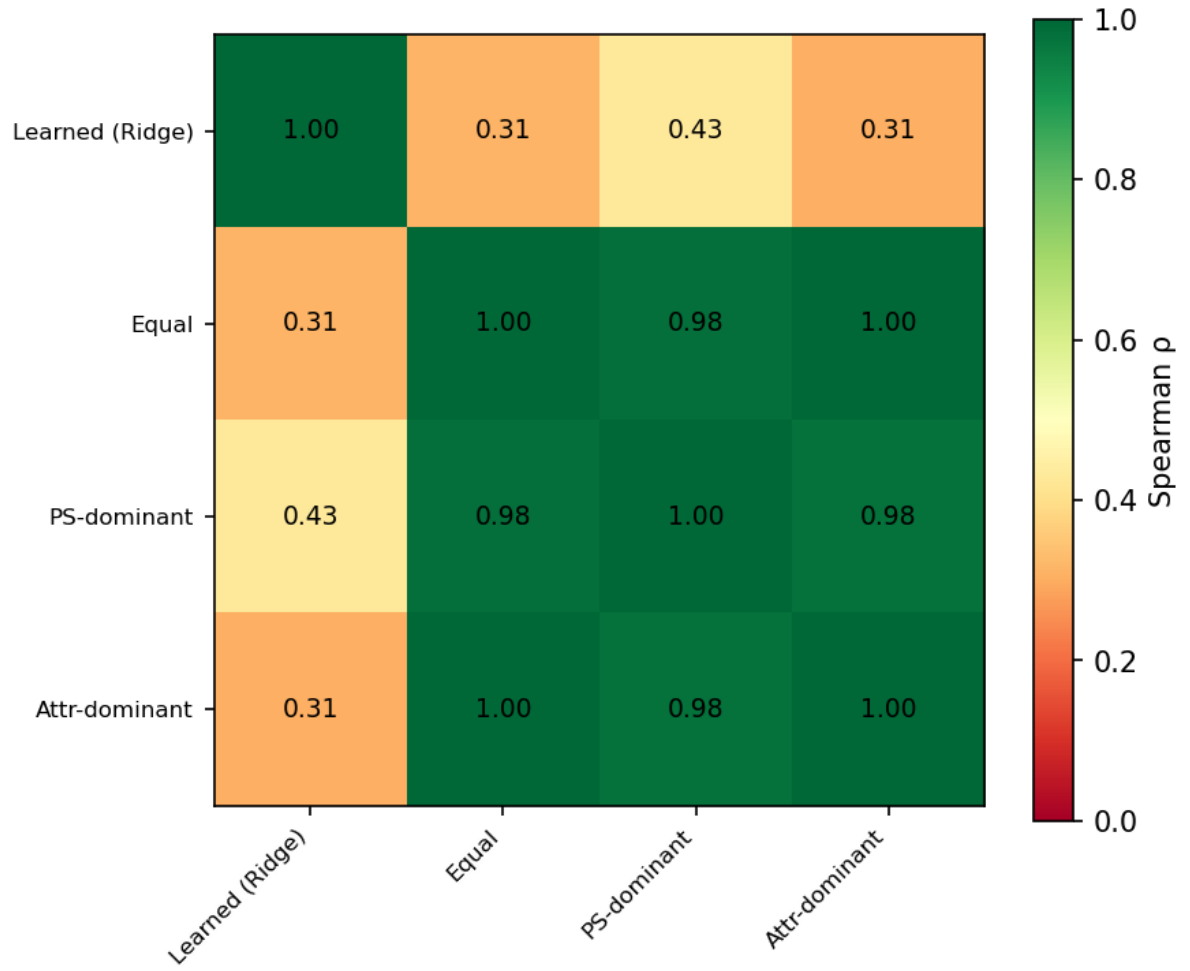


Figure 6: Spearman rank correlation across four weight schemes. Equal, PS-dominant, and Attr-dominant schemes produce near-identical rankings ($\rho > 0.97$). Learned (Ridge) scheme is qualitatively distinct.

Discussion

Key Findings

Finding 1: Responsibility concentration is a multi-layer computational signal. RCI integrates process, structural, and linguistic signals into a unified score, demonstrating that accountability concentration is detectable across representational layers. The framework provides a foundation for cross-layer accountability analysis in regulatory documents.

Finding 2: Process-structural signal dominates in FDA Warning Letters. The learned weight ($w_{ps} = 0.995$) reflects a corpus-structural property: FDA Warning Letters are issued to firms, not individuals, producing a structural pre-convergence in which accountability concentration is encoded in document architecture rather than linguistic choice. This finding has direct implications for accountability-aware AI design — systems that rely on linguistic signals alone will systematically underestimate concentration in enforcement documents.

Finding 3: WL2018-05 is weight-invariant. The extreme convergence case (single violation, zero explanatory diversity, pure Firm attribution) is identified as rank 1/50 across all weight schemes. This weight-invariance suggests that maximum accountability concentration produces a signal strong enough to dominate regardless of how layers are weighted.

Finding 4: Linguistic and process-structural layers are empirically independent. VIF = 1.03 confirms that the two compressed layers measure distinct dimensions of responsibility concentration. The near-zero learned weight for the linguistic layer does not reflect redundancy — it reflects the corpus-specific dominance of structural signals in FDA Warning Letters.

Finding 5: FDA Warning Letters exhibit structural pre-convergence. Firm attribution dominates (99% of actual labels) due to the enforcement logic of regulatory documents rather than investigative reasoning. RCI captures within-corpus variation despite this ceiling, but the ceiling itself is a finding: regulatory enforcement documents structurally pre-converge accountability before any investigative analysis occurs.

Limitations

Three limitations should be noted. First, the per-case signal reconstruction relies on outputs from three prior frameworks, each of which carries its own modeling assumptions and synthetic data components. Integration amplifies rather than cancels these assumptions. Second, the linguistic layer (attribution entropy) is computed from a reconstructed FDA Warning Letter violation text distribution rather than directly from case-level model outputs, introducing approximation error. Third, RCI is validated on a single corpus (FDA Warning Letters, 21 CFR 210/211); generalizability to other regulatory domains requires further validation with direct access to raw investigative data.

Conclusion

I presented the Responsibility Concentration Index (RCI), a multi-layer composite metric for accountability concentration in regulatory investigations. By integrating process-level PAC risk, structural centrality Gini, and linguistic attribution entropy into a single empirically weighted score, RCI operationalizes responsibility concentration as a quantifiable, cross-layer computational signal. Applied to FDA Warning Letters (N=50, 160 violations, 2016–2025), RCI identifies WL2018-05 as the extreme concentration case across all weight schemes, and reveals that process-structural signals dominate accountability concentration in regulatory enforcement documents ($w_{ps} = 0.995$) — a finding that reflects the structural pre-convergence of FDA enforcement logic rather than a limitation of the metric.

Future work will extend RCI to richer investigative corpora with direct access to in-process reasoning logs, incorporate transformer-based linguistic representations, and develop a multi-domain validation framework across healthcare, aviation, and financial regulatory environments.

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